

Assignment II – NS (CSC4065)

This assignment experiments with TLS certificates using the openssl toolkit, and gives a brief introduction to Wireshark. The openssl toolkit is commonly used to generate and check certificates, which will be used to encrypt communications and authenticate both ends. Some of the key tasks are as follows:

- generate a private RSA key and a certificate signing request (CSR),
- generate an x.509 certificate and learn to inspect the certificate,
- setup a simple TCP server with TLS, and
- use Wireshark to inspect the traffic between the server and the client.

Note: while it is not strictly necessary to use a Kali VM for this assignment, it is recommended to avoid further issues. Virtual Machines are disposable, so you can play with them and then delete them without worrying about making changes to your operating system or losing anything important.

Make sure that openssl and Wireshark are installed on your Kali VM (they should be pre-installed):

```
# Check if openssl is installed already
openssl version
# Check if wireshark is installed
wireshark --version
# OTHERWISE, install openssl and wireshark using your package manager
sudo apt-get update \
&& sudo apt-get upgrade -y \
&& sudo apt install openssl wireshark -y --fix-missing
```

You can find the exercises on the next page. Please be concise in your answers.

1. First, generate an RSA private key of 1024 bits and save it in a file. Then, generate a certificate signing request (CSR) using the RSA key you just created. You can populate various fields of the certificate with information regarding location, company, etc. Include the code used and the output generated and answer the following:
 - a. RSA with a key length of 1024 was used. Can you see any potential issues already?
 - b. The CSR was created from a private key. What would happen if this key is shared?
2. In general, there are two types of certificates: those signed by a trusted Certificate Authority (CA), and self-signed certificates. In this exercise, generate a self-signed certificate. You can use the common x.509 standard to generate a public key certificate, which will provide TLS encryption for the HTTPS server in the next exercises. Include the code used and answer the following:
 - a. What would happen if you connected to a server with expired certificates?
 - b. Can you see any potential issue with extending the validity of the certificate?
 - c. What about connecting to a server with a certificate generated using weak cryptographic functions?
3. Set up a test TCP server listening on port 1443 on your machine, using the certificate and private key you just generated in the previous exercises. You will likely get an error! Generate a new certificate using a longer key, for example RSA 4096, and start the server. Connect to the server from your preferred browser (still in the VM) on the address `https://localhost:1443`. What do you notice?
4. In this exercise, Wireshark will be used to inspect the traffic generated between the TCP server. Start capturing traffic on the `lo` interface (adapter for loopback traffic), where you will see all the requests are sent to the server (reload the page on your browser once or twice to see Wireshark getting populated with traffic traces). Then, answer the following questions:
 - a. Which version of TLS is being used?
 - b. Which symmetric encryption/authentication algorithm is used during the TLS negotiation between the server and the browser?
 - c. Navigate to <https://pwr.nu.edu.pk/computer-science/> and compare the packet headers with the ones from our server. Can you read the website content from the packet?
 - d. Use the keys generated during handshakes to decrypt TLS traffic in Wireshark and show your work. You can use this article¹ as a guide. Finally, explain why this is possible.

¹ [Decrypt SSL with Wireshark](#)